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Strategic Disclosure in Congested Low-Altitude Drone Traffic

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FINAL POSTER · VALIDATED BASELINE

KEY MESSAGE

“The largest benchmark coordination failure appears at intermediate congestion, where additional disclosure is socially valuable but remains privately unattractive to confidentiality-sensitive operators.”

STYLIZED
BENCHMARK

01 Question & model

Research question & verified gap

PRIMARY When operators privately differ in confidentiality cost, how does congestion affect strategic disclosure, and when does BNE fall below the full-information first best?

INTEGRATED How should low-altitude traffic systems structure non-safety sharing when confidentiality remains private but coordination is collective?

SETTING Capacity-constrained urban drone-delivery corridors. Stakeholders: operators, traffic authority, customers, nearby/public stakeholders, emergency users.

VERIFIED GAP Existing AAM work studies protocols, privacy mechanisms, and auctions. Here disclosure granularity is an endogenous choice under private commercial cost and congestion-dependent information value.

Strategic model & foundations

STATIC BAYESIAN GAME · INCOMPLETE INFORMATION

Players A, B. Private $c_i \in \{1, 1.6\}$; $\Pr(L)=\Pr(H)=0.5$. Public d: Low .3 | Medium .6 | High 1.0.

Actions: Minimum = mandatory/basic safety only; Partial = next sector + coarse ETA + traffic volume; Full = richer route/timing/trajectory.

Safety floor: Minimum never withholds required or safety-critical information.

$u_i = dB[1 - \exp(-\lambda(x_i + x_j))] - c_i k(a_i)$

$B=7.5$; $\lambda=.5$; $x=(0,1,2)$; $k=(0,1,2.5)$. Solution: pure-strategy Bayesian Nash equilibrium.

Three complementary lenses

GAME THEORY Private types + M/P/F disclosure → Bayesian equilibrium

SOCIAL CHOICE BNE welfare vs full-information first best → efficiency and burden

MECHANISM DESIGN CCAR access rule + priority allocation → incentive robustness

Downstream Week 5 allocation application

Resource: one commercial priority slot, $K=1$; emergency/public-safety missions remain outside the auction.

Values $A/B/C/D = 8/5/3/2$; FCFS arrival $B \rightarrow C \rightarrow A \rightarrow D$.

FCFS B wins | value 5 | efficiency 62.5%

SECOND PRICE A wins | payment 5 | utility 3 | efficiency 100%

Boundary: downstream scarcity allocation, not the disclosure mechanism.

02 Design & evidence

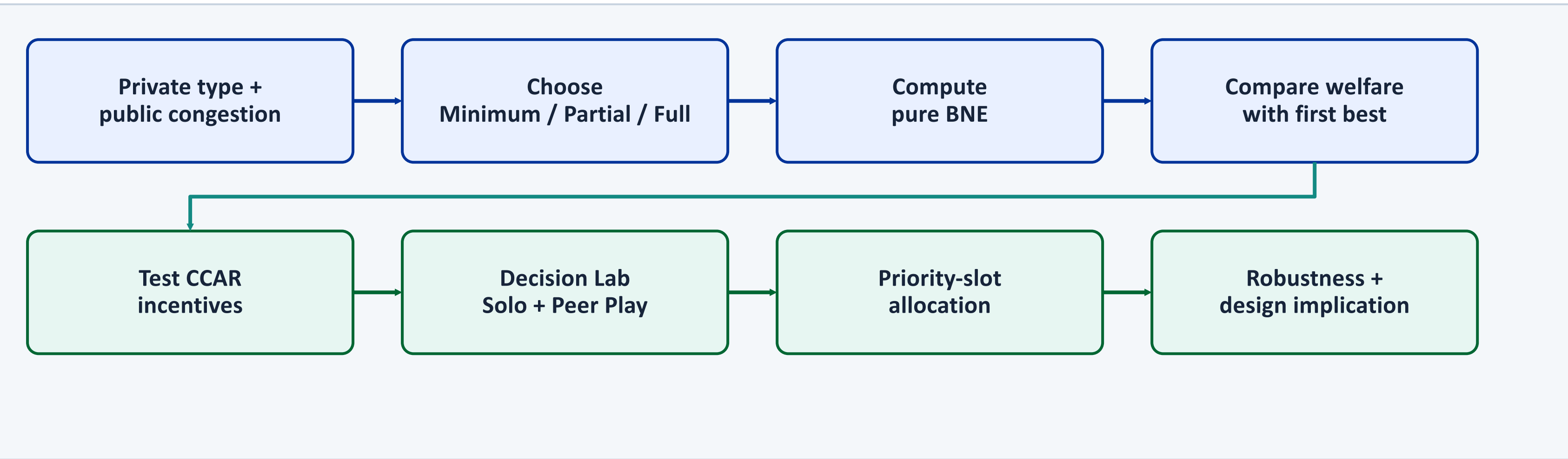
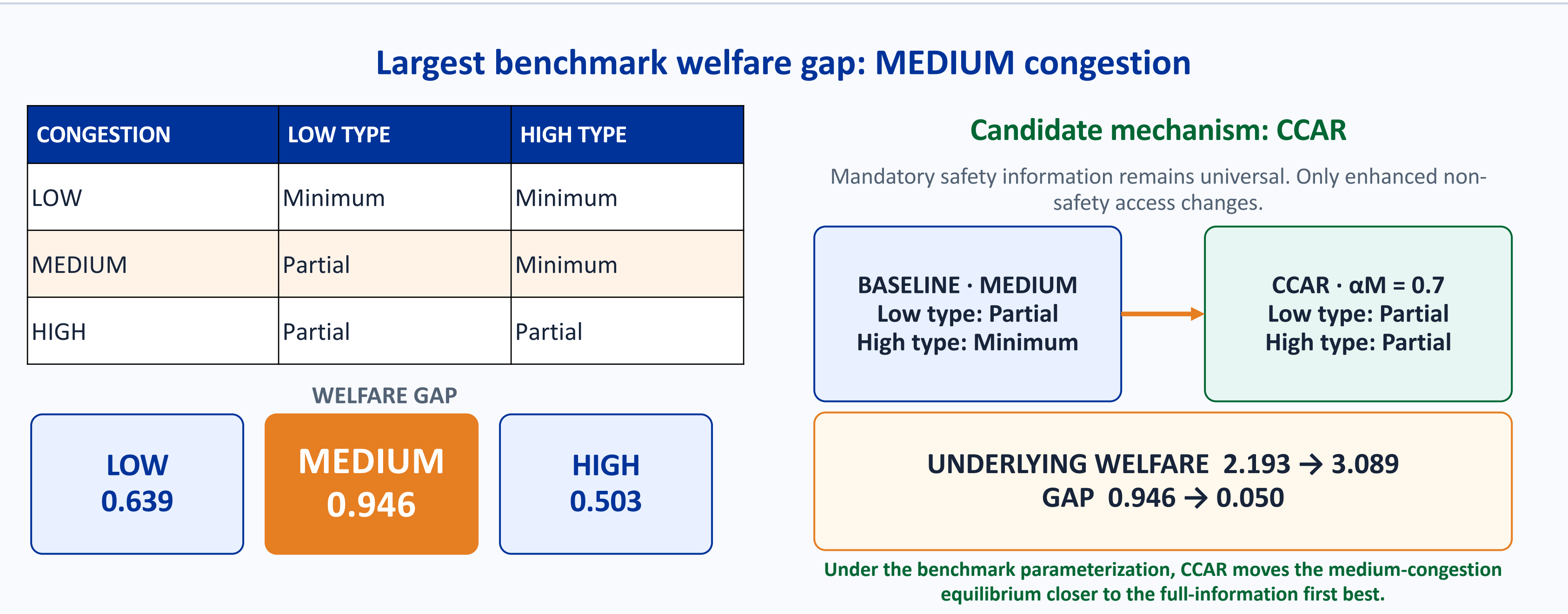


Figure 1. Economic model → verified computation → behavioral probe → bounded allocation → revision gate.



WHY MEDIUM? At high congestion, even the high-confidentiality type chooses Partial. At medium congestion, added disclosure is socially valuable but remains privately unattractive.

ROBUSTNESS 760 validated cells: 401 no equilibrium-set effect | 334 welfare improvement | 18 excessive-disclosure diagnostic | 3 larger gap | 66 multiple pure BNE | 0 no pure BNE. Categories overlap; these are not probabilities.

DESIGN MESSAGE Mechanism performance is parameter-dependent.

03 Impact & dialogue

Practical impact: a real-world case

CASE Commercial drone-delivery operators sharing capacity-constrained urban low-altitude corridors.

DECISION How much non-safety flight-intent information to share; how to allocate residual commercial priority scarcity.

POTENTIAL BENEFIT Better coordination, lower delay, more efficient corridor use.

RISKS Commercial confidentiality, unequal access, over-disclosure, misspecified incentives.

BEFORE ADOPTION Pilot calibration and confidentiality values; latency/reliability; safety and distributional review.

CORRECTION Auditable rule, safety-information floor, appeal/review, parameter revision, or suspension.

Interdisciplinary evidence

TESTABLE IMPLICATION

In an intermediate-congestion pilot with synthetic or calibrated confidentiality types, conditioning only enhanced non-safety coordination access on Partial/Full disclosure should increase Partial disclosure among confidentiality-sensitive operators relative to the baseline rule.

QUALIFY / REJECT IF disclosure becomes excessive; the welfare gap rises; access becomes systematically unequal; or operational/privacy costs dominate.

BEHAVIORAL SCIENCE Hugging Face Decision Lab: Solo Decision, reflection, Peer Play, benchmark reveal, CCAR comparison, auction exercise.

Evidence boundary: exploratory classroom decision exercise; not representative evidence about real drone operators.

COMPUTER SCIENCE GitHub + Colab 01 (BNE / welfare / CCAR) + Colab 02 (FCFS / second price).

Integration: the model defines the benchmark shown in the artifact; participant reflection identifies where predictions may require behavioral qualification.

Limitations & next test

LIMITS Two operators and types; independent private types; pure strategies; stylized parameters; no calibration; first best may be nonimplementable; CCAR cost omitted; IPV auction; interdependent mission values; exploratory demo.

NEXT TEST Compare baseline and CCAR under trusted-partner and competitive framing; classify deviations by privacy salience, reciprocity/fairness, or strategic misunderstanding.

OPEN DESIGN QUESTION At what congestion and confidentiality conditions should enhanced access become conditional without creating excessive disclosure or unequal access?

PEER REVIEW: PARTIAL

Superseded-version review only; no final-topic review.

Validated computational checkpoint: ffa4836208a3f647181771646c5639286c0726e6

Acknowledgments

Course instructor: Professor Luyao Zhang. No empirical operational dataset was used; inputs are stylized. Software: Python, NumPy, Matplotlib, nbclient, Gradio. Layout adapted from the official COMSCI/ECON 206 PS2 A0 Poster Template. Research question, model choices, interpretation, and final responsibility are the author's. AI tools assisted implementation, debugging, literature-search support, reproducibility checking, formatting, and drafting/editing.

Selected references

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OPEN MATERIALS

GitHub · code + reproducibility
HF · try the decision lab
Colab 01 · BNE / welfare / CCAR | Colab 02 · allocation
65 passed, 0 failed · hosted Run all confirmed

